

ENTS 656 Introduction to Cellular Communication Networks

Fall 2025 Project

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1. Introduction

1.1 Abstract

The primary objective of the project is to create a Python application that simulates the behavior of a single-cell wireless communication system. The simulation uses a discrete event modelling method and incorporates realistic channel effects such as path loss, shadowing and Rayleigh fading. The application then evaluates probabilities of a call getting blocked - either due to insufficient signal strength or due to channel unavailability as a function of number of users. This data is further used to determine the maximum number of users/mobile stations that can be supported while maintaining a Grade of Service (GOS) of 2%. The results demonstrate complex interplay between channel impairments, resource availability and the quality of service which helps in cellular network planning and optimization.

1.2 Problem Statement

The cellular system must support maximum user capacity while maintaining acceptable quality of service. The challenge lies in efficiently utilizing the limited resources in an environment with unpredictable propagation conditions.

The project generates two plots for the system, displaying the data of

1. probability of blocking vs number of users.
2. Probability of call drops vs number of users.

The project addresses the following questions:

1. How many MSs can this system serve with the given environment parameters and GOS?
2. How can the system be modified to improve its GOS?

3. How can the number of users be increased without increasing the number of available channels?

The analysis provides insight into which parameters dominate the performance of the system and whether the system is limited by coverage or channel capacity.

1.3 Report Organization

This report is organized as follows:

Section 2 details the system parameters used in the simulation: cell layout, base station characteristics, mobile station characteristics, traffic assumptions, quality of service statistics.

Section 3 describes channel model, including path loss, shadowing and fading.

Section 4 presents call state modeling and show how call generation, blocking, dropping and termination are handled.

Section 5 discusses structure of the python program, various modules and their purposes.

Section 6 reports simulation results and discusses impact of various system parameter modifications on GOS and user capacity.

Section 7 summarizes key finding and conclusions from the simulation.

2. System Parameters

2.1. The cell layout and Base Station (BS) parameters

The simulation models a single circular cell with the base station positioned at coordinates (0,0). Table 1 summarizes the key parameters of the cell.

Parameter	Variable name	Value	Unit
Cell Radius	cell_radius_km	1.5	km
Carrier frequency	carrier_f_MHz	900	MHz
BS transmit power	P_Tx_W	10	W
BS feeder loss	Feeder_loss_Tx	4	dB
BS antenna gain	Antenna_Gain_Tx	8	dB
BS antenna height	Antenna_h_Tx	30	meters
Max available channels	Total_Channel_Count	50	-

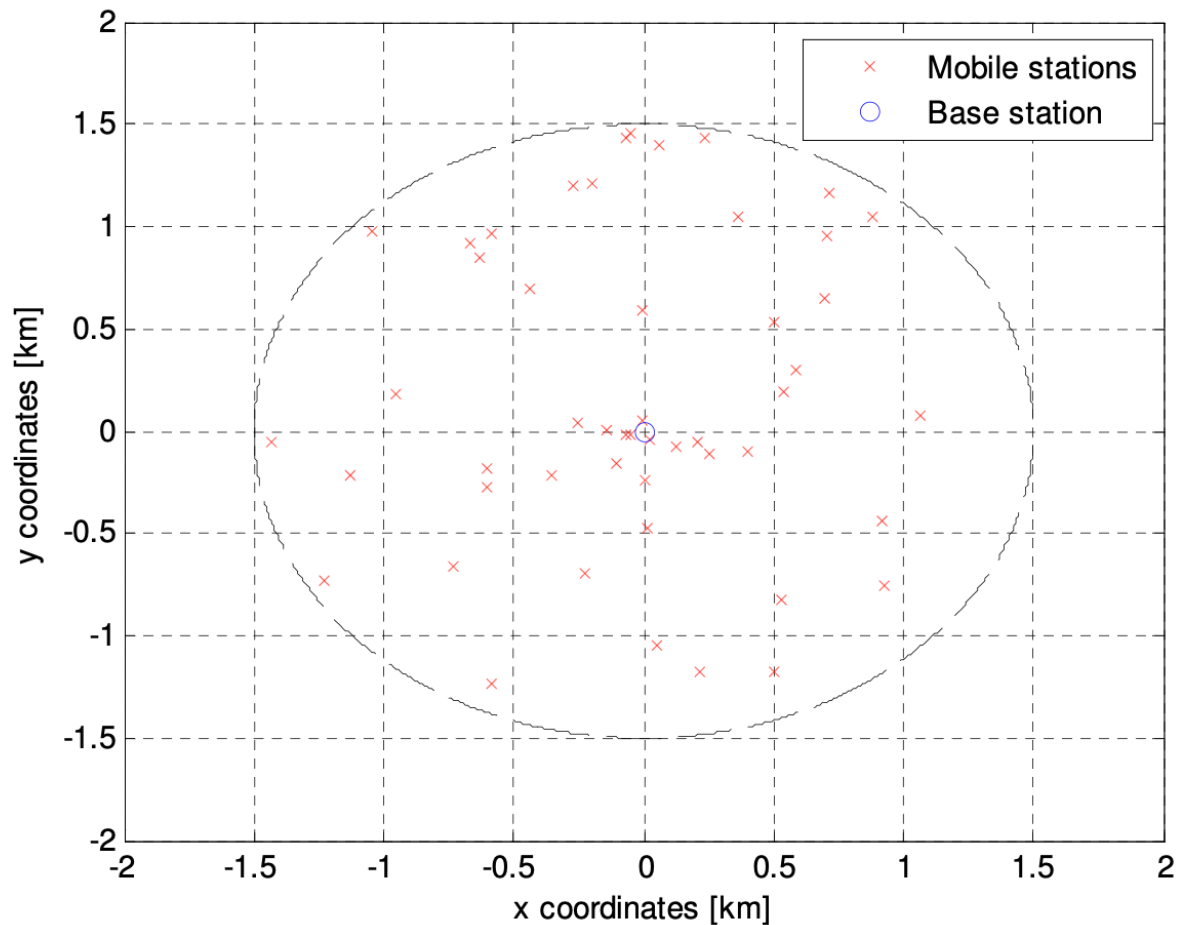
2.2. Mobile Station (MS) parameters

Mobile stations properties are as below.

Parameter	Variable name	Value	Unit
Receiver sensitivity	P_Sensitivity	-102	dBm
Orientation loss	Orientation_loss_Rx	3	dB
BS antenna gain	Antenna_Gain_Rx	3	dB
BS antenna height	Antenna_h_Rx	1.5	meters

Signal power below the receiver sensitivity of -102 dBm will result in blocked or dropped calls.

MS positions are modeled using distance (radius) and angle from BS. Proper randomness is ensured by using uniform function on radius^2 and angle. This is handled by the function `MS_location_assignment()` in `propagation_loss_functions.py` file.



2.3. Call traffic model

This section deals with parameters which govern the call generation and call duration.

Parameter	Variable name	Value	Unit
Call request per user per minute	lambda_call_request_per_user_per_min	0.017	Calls/minute
Call duration	call_duration_min	3	minutes

For given discrete time interval (modeled with variable $dt = 1 \text{ second}$), the probability of call request generated by an idle MS is:

$$\text{Probability (call generation)} = (\text{Call request per user per minute}) * dt / 60$$

The call is terminated gracefully after 3 minutes of duration. This termination is not pegged as an error.

2.4. Quality of Service parameters

The simulation tracks three quality of service metrics:

probability of blocked calls due to channel unavailability	p_bc
probability of blocked calls due to weak signal strength	p_bs
probability of dropped calls	p_d

1. Probability of blocked calls due to channel unavailability (p_bc):

$$P_{\text{block_channel}} = (p_{bc}) = \frac{\text{Total Blocked Calls due to channel}}{\text{Total Calls Generated}}$$

2. Probability of blocked calls due to insufficient received signal strength at MS (p_bs):

$$P_{\text{block_signal}} = (p_{ds}) = \frac{\text{Total Blocked Calls due to signal}}{\text{Total Calls Generated}}$$

3. Probability of dropped calls due to channel unavailability (p_d): The ratio of successful calls that got terminated because the signal strength dropped below receiver sensitivity ($P_{\text{Sensitivity}}$) due to deep fading.

$$P_{\text{dropped}} = (p_d) = \frac{\text{Total Dropped Calls}}{\text{Total Successfully Initiated Calls}}$$

System capacity is determined by the maximum number of users that satisfies the given GOS requirement.

$$P_{\text{block_channel}} + P_{\text{block_signal}} \leq \text{GoS}$$

The given GOS value is 0.02

3. Channel Propagation Modeling

The received power at the MS is calculated in dBm using the link budget equation:

$$P_{\text{Recieved_Power}} = P_{\text{Transmitted_Power}} - L_{\text{BS_Feeder_Loss}} + G_{\text{TX_Antenna_Gain}} + G_{\text{RX_Antenna_Gain}} - L_{\text{Orientation_Loss}} - L_{\text{Hata_Loss}} - L_{\text{Shadowing}} + G_{\text{Fading}}$$

The channel model is a composite of three major effects - Path Loss, Shadowing, and Fading - which has been modelled in different functions as explained below.

3.1 Path loss

The Path Loss is determined using Okumura-Hata model.

$$L_{\text{Hata}}(d) = 69.55 + 26.16 \log_{10}(f_c) - 13.82 \log_{10}(h_{TX}) - [3.2(\log_{10}(11.75h_{RX}))^2 - 4.97] + (44.9 - 6.55 \log_{10}(h_{TX})) \log_{10}(d)$$

Here d is the distance of MS from BS in km. f_c is the frequency of the carrier wave in MHz. h_{TX} is the height of the transmitter antenna and h_{RX} is the height of the receiver antenna.

This calculation is performed in the function named Path_loss_Hata() in file propagation_loss_functions.py

3.2 Shadowing

Shadowing accounts for obstructions along the path from transmitter to receiver like buildings and hills. It is modeled using Log-Normal random variable. In dB, this will convert into Gaussian Normal random variable with zero mean and a standard deviation of 5 dB.

This calculation is performed in the function named `Path_loss_shadowing()` in file `propagation_loss_functions.py`

3.3 Fading

Fading models rapid fluctuations in signal strength due to multipath propagation. As per the requirement of the project, the instantaneous fading is calculated every 1s of time. The effect is modeled as Rayleigh fading.

For calculation, the absolute value of complex channel coefficient h (where $h = x + jy$) is generated using variables x and y which are independent and distributed as $N(0, \frac{1}{2})$. The variance of individual x and y as $\frac{1}{2}$ is arrived at by dividing Rayleigh fading unit variance by 2. The linear gain in dB is obtained from this amplitude using $20\log_{10}$ formula.

This calculation is performed in the function named `Fading_effect()` in file `propagation_loss_functions.py`

4. Call State Modeling

Each MS has two primary states:

Idle state – where no call is ongoing represented by `call_ongoing` variable with value 0,

Call ongoing state – where MS is in an ongoing call represented by `call_ongoing` variable with value 1.

The state transitions and relevant conditions are explained in flow diagram below.

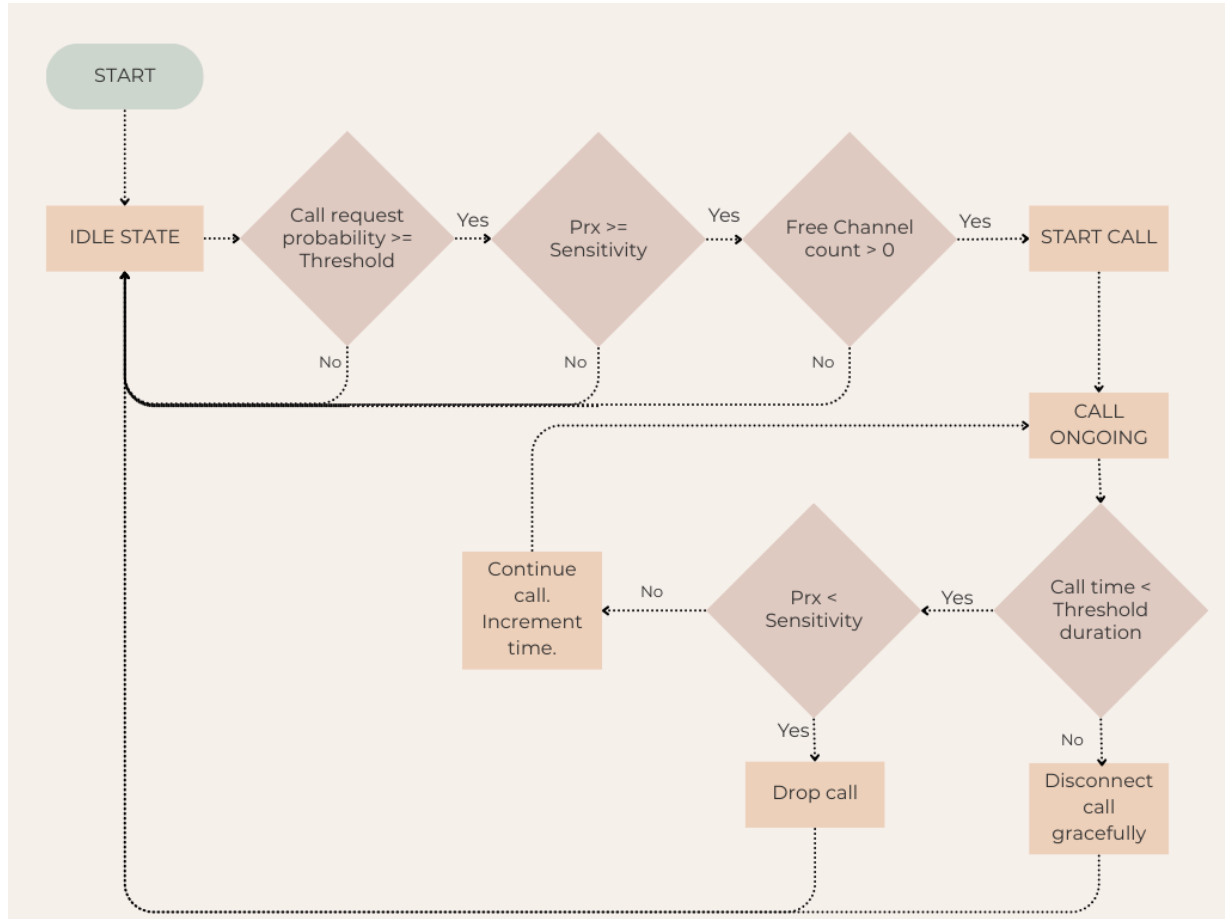


Figure 1 Call state modeling

5. Design and Flow of the Simulation

The simulation is run multiple times (controlled by `simulation_count` variable) for each user count (`num_of_users_list`). In each iteration, the set of MS locations, Hata path loss and shadowing loss are fixed while the fading effect is recalculated every 1s (controlled by variable `dt`) time calculated in the simulation. Every `dt` time, the calls are generated, tracked and controlled based on the given parameters.

The system consists of the following four main modules:

1. Parameter configuration module (`config.py`): This module holds the system constants such as cell radius, carrier frequency, transmit power, antenna height and other parameters.
2. Propagation module (`propagation_loss_function.py`): This module has functions that computes path loss, shadowing, fading as well as the function to generate random positions for various MSs in the cell.

3. Simulation module (simulation_functions.py): This module has functions that calculates received power per user. It also generates, tracks, blocks and drops calls based on various observations like probability, signal strength and calculates the Key Performance Indicators (KPIs) of the system.
4. Main module (main.py): This module generates visualizations and performance metrics and calls on the functions to create end-to-end simulation of the system.

6. Simulation results and Suggestions

The system is evaluated for user counts starting with 10 under baseline configurations. Key performance indicators were recorded to determine whether the system satisfies the GOS requirement.

Baseline Simulations

The simulation started with number of users at 10 to 50. The output showed minimal blocking due to lack of channels, while the blocking due to lack of received signal power causes the GOS to breach the given value.

User Count	P(blocked:signal)	P(blocked:channel)	P(blocked:total)	P(dropped)
10	0.024	0.0	0.024	0.597
20	0.033	0.0	0.033	0.764
30	0.027	0.0	0.027	0.676
40	0.020	0.0	0.020	0.632
50	0.024	0.0	0.024	0.693

This shows that the given GOS is not attainable for the system with these baseline parameters for any number of users since the received signal power is too low (coverage limited).

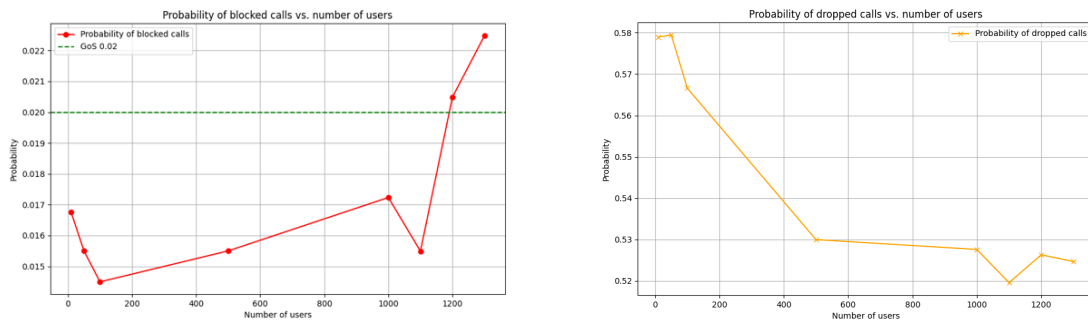
Possible Modifications to the System

To improve the system performance, a series of modifications were applied and evaluated. A few changes were also applied independently to show its standalone impact to the system.

Change 1: Increase the transmit signal power to from 10W to 20W.

User Count	P(blocked:signal)	P(blocked:channel)	P(blocked)	P(dropped)
10	0.017	0.0	0.017	0.579
50	0.015	0.0	0.016	0.579
100	0.014	0.0	0.015	0.567
500	0.015	0.0	0.016	0.53
1000	0.017	0.0	0.018	0.528
1100	0.014	0.001	0.016	0.52
1200	0.017	0.003	0.021	0.526
1300	0.017	0.006	0.023	0.525
Given system can provide GoS 0.02 for 1100 users				

Figure 2 User count vs probability of block/drop for change 1



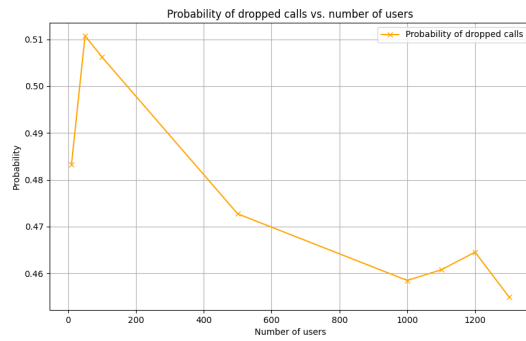
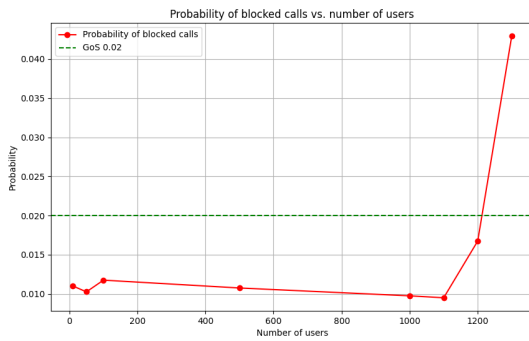
Now the simulations show that the system can support around 1100 users, as probability of blocked calls due to unreliable signal reduced substantially. This is because improved transmit power automatically converts to improved received power.

Change 2: Additionally, increase the BS height from 30m to 40m

As shown in table below, probability of blocked calls due to signal has reduced further.

User Count	P(blocked:signal)	P(blocked:channel)	P(blocked)	P(dropped)
10	0.011	0.0	0.011	0.483
50	0.01	0.0	0.01	0.511
100	0.012	0.0	0.012	0.506
500	0.011	0.0	0.011	0.473
1000	0.01	0.0	0.01	0.459
1100	0.009	0.0	0.01	0.461
1200	0.01	0.007	0.017	0.465
1300	0.011	0.031	0.045	0.455
Given system can provide GoS 0.02 for 1200 users				

Figure 3 User count vs probability of block/drop for change 2



With this additional change, the system can support 1200 users. This improvement can be understood by looking closely at the Okumura-Hata equation which shows that the loss in power is reduced when BS antenna height is increased.

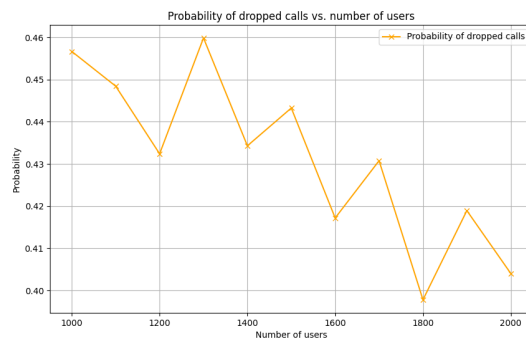
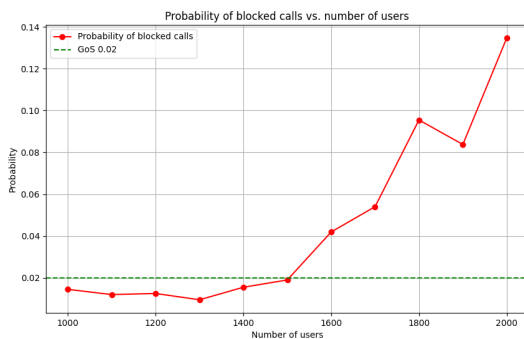
Further improvements to the received signal cannot improve the supported user count considerably as GOS is now limited by number of available channels.

Change 3: After above changes if we increase the channel count from 50 to 60, the system can support 1500 users as more channels are now available for simultaneous calls of multiple users.

User Count	P(blocked:signal)	P(blocked:channel)	P(blocked)	P(dropped)
1000	0.014	0.0	0.015	0.457
1100	0.012	0.0	0.012	0.448
1200	0.013	0.0	0.013	0.432
1300	0.009	0.0	0.01	0.46
1400	0.008	0.007	0.016	0.434
1500	0.011	0.007	0.019	0.443
1600	0.013	0.028	0.044	0.417
1700	0.011	0.043	0.057	0.431
1800	0.012	0.083	0.106	0.398
1900	0.007	0.076	0.091	0.419
2000	0.011	0.123	0.156	0.404

Given system can provide GoS 0.02 for 1500 users

Figure 4 User count vs probability of block/drop for change 3



NOTE – Below changes are performed from baseline configurations, to see the impact of the change without cumulative effect of other changes.

Change 4: Increase BS antenna gain from 8dB to 12dB

User Count	P(blocked:signal)	P(blocked:channel)	P(blocked)	P(dropped)
1000	0.009	0.0	0.01	0.488
1100	0.012	0.002	0.014	0.477
1200	0.016	0.005	0.022	0.476
1300	0.011	0.009	0.02	0.476
1400	0.012	0.027	0.042	0.473
1500	0.011	0.054	0.069	0.486
1600	0.018	0.067	0.094	0.491
1700	0.012	0.115	0.146	0.447
Given system can provide GoS 0.02 for 1300 users				

Figure 5 User count vs probability of block/drop for change 4

System supports 1300 users with this configuration change alone. This change causes improvement in the received signal as overall power of signal is improved. Thus, more users can receive the signal with sufficient strength.

Change 5: Decrease the radius covered by cell from 1.5km to 1.2km

User Count	P(blocked:signal)	P(blocked:channel)	P(blocked)	P(dropped)
1000	0.014	0.0	0.014	0.509
1100	0.014	0.001	0.015	0.508
1200	0.016	0.0	0.017	0.489
1300	0.014	0.004	0.019	0.491
1400	0.016	0.014	0.031	0.502
1500	0.011	0.041	0.056	0.494
1600	0.011	0.07	0.089	0.49
Given system can provide GoS 0.02 for 1300 users				

Figure 6 User count vs probability of block/drop for change 5

System supports 1300 users with this configuration change alone. This will exclude area further away from the cell which receive weaker signals from the cell. As these contribute largely to the deterioration of statistics, excluding them improves the GOS. The users in the excluded area can be served by a different cell.

7. Conclusions

This project developed a Python based simulator to analyse performance of a single-cell wireless communication system under realistic propagation conditions by modeling path loss, shadowing, fading, user behavior and channel allocation to provide insights into the relationship between various system parameters and quality of service.

The baseline configurations were found to not be sufficient to run a system with GOS of 2%. The received signal power frequently fell below the sensitivity threshold.

Several system modifications were evaluated to improve performance of the system. Increasing the transmit power and antenna height significantly enhanced coverage and allowed the system to support 1200 users with GOS of 2%. With these changes the system turned from coverage limited into channel capacity limited system.

The channel count was found to restrict the system to support around 1200-1300 maximum users. Further scaling the channel count from 50 to 60, enabled the system to support 1500 users.

The simulations also showed improvement in number of users by independently increasing antenna gain values and decreasing cell radius.

Overall, the results demonstrate impact of capacity constraints (channel constraint) and impact of radio parameters (power, antenna height, antenna gain, cell radius) in providing adequate GOS to a system. The simulation successfully captures these complex interactions and provides assistance in cellular network planning and optimization.

***** THE END *****